

BALA SAI IRRINKI

DATA ANALYST | DATA ENGINEER

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CAREER SUMMARY

Results-driven Data Engineer and Data Analyst with 5+ years of experience designing scalable data pipelines, cloud data warehouses, and analytics solutions across financial services and enterprise domains. Expert in Python, SQL, PySpark, and the modern data stack including Apache Airflow, dbt, Snowflake, and Databricks with hands-on experience across AWS and Azure cloud platforms. Proven ability to architect end-to-end ETL/ELT pipelines, build real-time streaming architectures using Apache Kafka, and deliver executive-ready dashboards and analytical insights using Tableau and Power BI. Strong track record of bridging data engineering and data analytics building the infrastructure that powers insights while also delivering the insights themselves. Adept at enforcing data quality, governance, and compliance standards in highly regulated financial services environments.

TECHNICAL SKILLS

- **Programming & Query Languages:** Python (Pandas, NumPy, PySpark, scikit-learn), SQL (advanced CTEs, window functions, stored procedures, query optimization), R, Scala (reading), Shell/Bash, DAX, MDX
- **Data Engineering & Pipelines:** Apache Airflow, dbt (data build tool), Apache Spark, PySpark, Apache Kafka, Apache Flink, Spark Streaming, AWS Glue, Azure Data Factory, Apache NiFi, Fivetran, Informatica, SSIS, ETL/ELT Pipeline Development, Data Ingestion, Batch Processing, Real-Time Streaming
- **Cloud Platforms & Services:** AWS (S3, EC2, Lambda, Redshift, Kinesis, EMR, Glue, SageMaker, RDS, Athena), Azure (Azure Data Lake Storage Gen2, Azure Synapse Analytics, Azure Data Factory, Azure Databricks, Azure ML, Event Hub), Google Cloud Platform (BigQuery, Dataflow, Cloud Storage)
- **Data Warehouses & Lakehouses:** Snowflake, Databricks (Delta Lake), Amazon Redshift, Azure Synapse Analytics, Google BigQuery, Microsoft Fabric, Lakehouse Architecture, Medallion Architecture (Bronze-Silver-Gold), Data Mesh
- **Databases:** PostgreSQL, MySQL, SQL Server, Oracle, MongoDB, DynamoDB, Cassandra, Redis, Elasticsearch
- **Data Analysis & Visualization:** Tableau (LOD expressions, calculated fields, story points), Power BI (DAX, Power Query, paginated reports), Looker, Excel (advanced formulas, pivot tables, VBA), Matplotlib, Seaborn, Plotly
- **Statistical & Analytical Methods:** Hypothesis Testing, A/B Testing, Regression Analysis (linear, logistic, multivariate), Time Series Forecasting (ARIMA, Prophet), Cohort Analysis, RFM Analysis, Customer Segmentation, Statistical Modeling, Predictive Analytics
- **Machine Learning & AI:** scikit-learn, XGBoost, Random Forest, Gradient Boosting, K-Means Clustering, Feature Engineering, MLflow, AWS SageMaker, Azure ML, LangChain, RAG Pipelines, LLM Data Infrastructure, Embedding Pipelines, AI-Assisted Analytics, GitHub Copilot
- **Data Quality & Governance:** dbt Tests, Great Expectations, Monte Carlo, Apache Atlas, Data Lineage, Data Cataloging, Data Profiling, Master Data Management, GDPR, SOX, PCI DSS, HIPAA
- **DevOps & Infrastructure:** Docker, Kubernetes, Terraform, Git, GitHub Actions, Jenkins, CI/CD Pipelines, Infrastructure as Code (IaC), DataOps
- **Methodologies:** Agile/Scrum, JIRA, Confluence, Data Modeling (Star Schema, Snowflake Schema, SCD Type 1/2), Dimensional Modeling, Data Storytelling, Stakeholder Communication

PROFESSIONAL EXPERIENCE

DATA ENGINEER | PNC Bank - Texas, USA

September 2023 – Present

- Design and implement end-to-end ELT pipelines using Azure Data Factory, PySpark, and Databricks ingesting 300M+ daily financial transactions and banking records into a Medallion (Bronze-Silver-Gold) lakehouse architecture on Azure Data Lake Storage Gen2, improving data freshness from daily-batch to near real-time and reducing downstream reporting errors by 60%
- Build and maintain Snowflake data warehouse with optimized dimensional data models (star schema, SCD Type 2), clustering keys, and materialized views supporting 150+ analysts and business stakeholders across retail banking, risk, and compliance teams, reducing average query latency by 50% and lowering compute costs by 28% through warehouse right-sizing and workload optimization
- Develop and manage 70+ dbt models with full data lineage tracking, automated testing (not_null, uniqueness, referential integrity, custom business rule assertions), and self-serve documentation published to dbt Cloud, achieving 99.2% data quality scores across critical transaction and customer datasets
- Orchestrate 100+ production Apache Airflow DAGs for batch ingestion, data transformation, and ML feature extraction pipelines with automated alerting, SLA monitoring, and retry logic, maintaining 99.7% pipeline uptime and supporting on-time delivery of regulatory and business-critical reporting workflows
- Build real-time data streaming pipelines using Apache Kafka and AWS Kinesis consuming live transaction event streams, enabling sub-5-minute data latency for fraud detection dashboards and risk monitoring systems that protect 2M+ active banking customers across PNC's Texas market
- Develop and deploy 12+ automated executive dashboards and self-service analytics reports in Tableau and Power BI (DAX, Power Query) integrated with Snowflake and Azure Synapse, reducing manual reporting effort by 45 hours weekly and improving data accuracy to 99.5% through automated ETL validation

- Partner with data science and risk teams to engineer ML-ready feature stores in Databricks for credit risk scoring, fraud detection, and customer churn prediction models deployed via Azure ML, reducing feature engineering cycle time by 45% and supporting SOX and PCI DSS compliance requirements
- Implement data governance framework using Apache Atlas and dbt Tests, enforcing column-level data lineage, data quality contracts, and access controls across all PII and sensitive financial datasets in alignment with SOX, PCI DSS, and internal audit requirements, achieving zero compliance violations across production environment
- Build GenAI data infrastructure using LangChain and Azure OpenAI to support internal knowledge retrieval applications, designing RAG (Retrieval-Augmented Generation) pipelines that chunk, embed, and index structured and unstructured banking documents into vector databases for intelligent document search and automated report summarization
- Collaborate with cross-functional Agile teams of data scientists, analysts, product managers, and compliance stakeholders to deliver 3-4 major platform features per sprint, perform A/B testing and statistical analysis on product experiments, and translate complex data findings into executive-ready business narratives that inform strategic decision-making

DATA ANALYST | Dataone Solutions, INDIA

January 2020 – June 2022

- Architected and delivered scalable ETL pipelines using Python, PySpark, and Apache Spark on AWS (EMR, S3, Glue) for 8+ enterprise clients across healthcare, insurance, retail, and financial services domains, processing 100GB+ daily data volumes with 99.5% pipeline reliability and consolidating 12+ disparate source systems into unified analytical data platforms
- Designed and built cloud data warehouse solutions on Amazon Redshift and MySQL with dimensional data models (star schema, SCD Type 2) and automated ingestion workflows connecting to Tableau and Power BI dashboards, enabling self-service analytics for 80+ business users and reducing client reporting cycles from weekly to daily
- Developed healthcare analytics solutions for 3 major clients using Python (Pandas, scikit-learn) and SQL, building predictive models for patient readmission risk and treatment pattern analysis that reduced average patient stay by 2.1 days and decreased operational costs by \$3.8M annually through data-driven care pathway optimization
- Built insurance risk scoring models using logistic regression, decision trees, and Random Forest in Python for 5+ insurance clients, improving claim prediction accuracy by 32% and reducing fraudulent claim exposure by \$7.2M annually through advanced feature engineering and ensemble modeling on historical claims data
- Implemented customer segmentation, cohort analysis, and RFM modeling for 10+ e-commerce clients using K-Means clustering and Python, increasing customer lifetime value by 25% and improving conversion rates by 14% through data-driven targeted marketing strategies and behavioral funnel analytics
- Built time series forecasting models using ARIMA and Prophet algorithms for retail clients, achieving 87% forecast accuracy, reducing inventory carrying costs by 20%, and minimizing stockout incidents by 42% through automated demand planning pipelines integrated with supply chain reporting systems
- Orchestrated 50+ Apache Airflow DAGs managing multi-dependency data pipeline workflows with SLA monitoring, error handling, and automated email alerting, ensuring on-time data delivery for business-critical reporting and reducing manual pipeline intervention by 75%
- Designed and deployed 20+ interactive analytics dashboards across Tableau and Power BI for healthcare, insurance, and retail clients, enabling real-time operational decision-making and reducing manual reporting effort by 30+ hours weekly through automated data refresh and self-service analytics capabilities

EDUCATION

MASTER OF SCIENCE IN COMPUTER SCIENCE | University of Texas at Arlington, USA

BACHELOR OF TECHNOLOGY IN COMPUTER SCIENCE | CMR College of Engineering & Technology, India